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Cache Memory

Rate at which data can be fetched from memory is about 10 times slower than rate at which CPU can process data.

- extremely fast, small memory b/w CPU and main memory whose access time is closer to processing speed of CPU. It acts as a high speed buffer
- is used to — temporarily store very active data and instructions during processing.
- since it is fast, processing speed is improved by making data and instructions needed for current processing available in cache.

The speed of processor is directly related to computer's clock speed. which is no. of pulses produced per second. This clock speed is measured in megahertz (MHz) or gigahertz (GHz)

Registers are part of CPU (not main memory)

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Bus

↳ refers to collection of wires through which data is transmitted from one part of a computer to another.
↳ it is a communication channel.
↳ characteristic of bus is shared transmission medium.
↳ limitation of bus is only one transmission at a time.

A computer bus is divided into two types

(i) Internal Bus.
↳ used to provide communication b/w CPU, memory and I/O (input and output) is called system bus.

(ii) External Bus.
↳ it connects different external devices, peripherals, expansion slots, I/O ports and drive connections to rest of computer. Also called Expansion Bus.

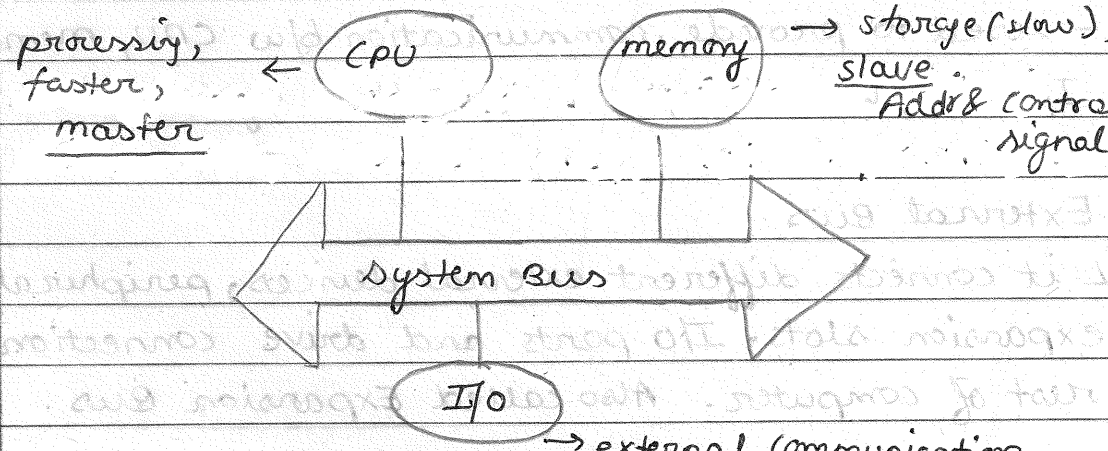
Some commonly used buses in microprocessor.

(i) Data Bus - These lines are used to carry data/binary sequence b/w CPU, memory, I/O. These lines are bidirectional. Memory is byte addressable (by default).

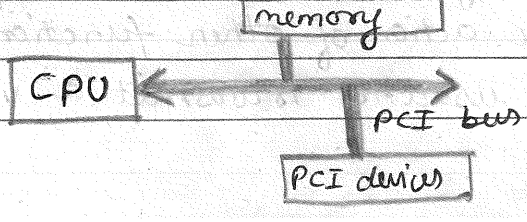
(ii) Address Bus - transfers information about where data should go. They are used to carry address towards memory & I/O. They are unidirectional. Based on address lines, we can determine capacity of memory system.

(iii) Control Bus - These lines are used to carry control signal and timing signal. It is used by CPU to direct and monitor action of other functional area of computer. It is used to transmit a variety of

individual signals (read, write, interrupt etc) to control and coordinate operation of computer.
We cannot say they are unidirectional / bidirectional. They are ON/ OUT ie IN unidirectional / OUT unidirectional.
↳ control signals are used to indicate type of operation whereas timing signal are used to synchronise memory and I/O operation with processor clock.



→ Peripheral component Interconnect (PCI)
(bus which attaches peripheral components) → like mouse, keyboard etc.
- high performance bus that is used to integrate chips, processors, memory sub-systems and expansion boards.
- also supports peripherals and devices like hard disk drives, networks etc.
- only bus to carry 64 bits of data in each clock cycle.
- developed by Intel Corporation.
- PCI successfully replaced all other older buses like ISA, EISA and VL.



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Advantages of PCI

(i) You can have multiple PCI buses on same computer.

(ii) You can connect maximum of five component to PCI and can replace them by fixed device on motherboard.

In short, system bus → connects CPU, memory and I/O.
PCI → interconnection b/w peripheral components. (ie I/O devices etc)

Input - Output Modules.

↳ connected with I/O devices and to system bus. ie normally connected to computer system on one end and one or more input/output devices on other.

I/O devices → I/O module → system bus → CPU/memory.

↳ I/O module is mediator b/w processor and I/O devices.
↳ Since peripherals are electromagnetic devices and data transfer rate of peripherals is usually slower than transfer rate of CPU (electronic devices), a conversion of signals value may be required ie. a synchronization mechanism may be needed.
↳ data format and word length used by peripheral may be quite different than that of CPU.
↳ It should not only communicate information from CPU to I/O devices but also coordinate them.

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Input Output Techniques.

(I) Programmed I/O.

→ Data transfer takes place under the direct control of processor.

→ processor must continuously check an I/O device and hence it cannot do another task. Hence this method is slow.

Characteristics of Programmed I/O.

(I) I/O operations are completely controlled by CPU.

(II) used in real time and embedded systems.

(III) used in CPUs, which have single input & output instruction. Each of these instructions selects one device.

(IV) **disadvantage:** CPU spends most of its time waiting for device to become ready.

(II) Interrupt Driven I/O.

In this, interrupt is used for any event that causes temporary transfer of control of CPU from one program to other which is causing interrupt.

Interrupts are primarily issued on

(I) Initiation of I/O operation (interrupt issued by I/O devices)

(II) Completion of I/O operation.

(III) occurrence of hardware or software errors.

How does it work?

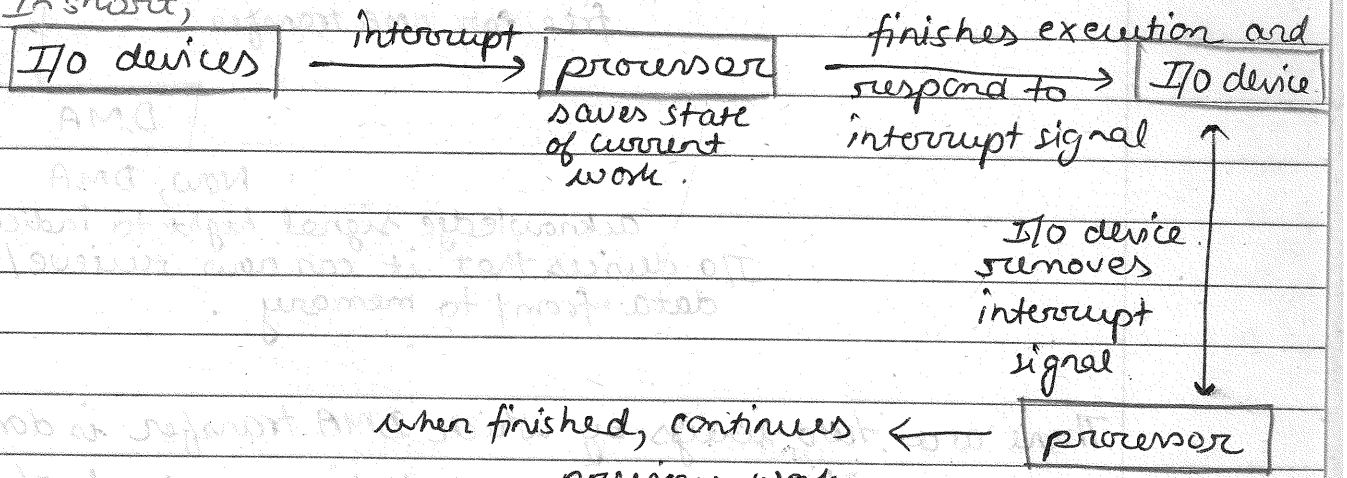
(I) In interrupt driver I/O, processor (CPU) issues a READ/WRITE instructions to device and then continue its work.

(II) When interface buffer (high speed buffer) is full, it is ready to send data to processor,

then interface sends a signal to processor (CPU) informing data is ready. This signal is called interrupt signal.

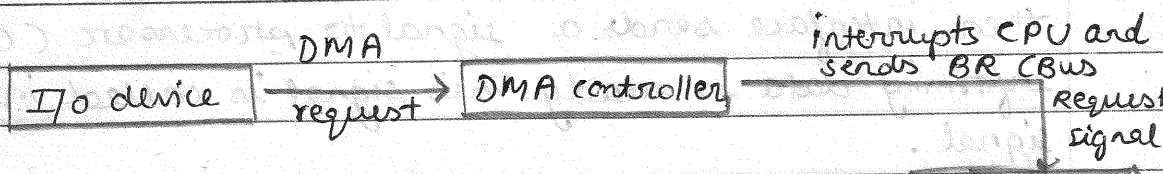
(iii) When processor receives interrupt signal. (ie. data is ready) it suspends its current job and transfers data from buffer to its own register.

In short,



disadvantage:
processor must suspend its work and later resume.
If there are many devices sending interrupt, then processor must be able to attend each of these based on some priority.

(iii) DMA: Direct Memory Access.
- method is required to eliminate need for continuous involvement of processor in I/O operation.
- we can facilitate I/O device to directly communicate with memory and keep CPU free for other task. For this purpose, DMA device is used.
- DMA controller is a device that facilitates data b/w memory and I/O device directly without interrupt of CPU.



CPU responds with BG (Bus Grand) lines that bus are free for DMA transfer

DMA

Now, DMA acknowledge signal high to indicate I/o devices that it can now recieve/send data from/to memory.

There are two ways by which DMA transfer is done:

(i) cycle stealing : process of taking control of memory cycle to transfer data. DMA transfer one data word at a time after which it must return control of buses to CPU. (ie asks again & again CPU) The CPU delays its operations for 1 cycle to allow DMA to "steal" one memory cycle.

(ii) Bulk transfer: all data is transferred after DMA acknowledgement. (All data is transferred at once and asks CPU only one time).

Input/output (I/O) channel provides communication b/w CPU and peripheral devices

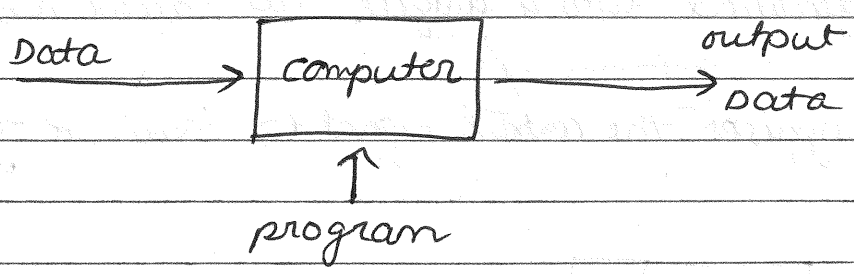
Simplex channel
communication is one direction only.
ie. radio

Duplex channel.
two way communication, one is for sending, one for receiving.
ie. telephone.

Dt.: Pg.: Delta

Functionality of computer system

It is a computational device used to process the data under control of program



While executing program on computer system, the input data will be processed into desired format therefore computer functionality is program execution

program is a sequence of instructions alongwith data.

program — instruction
program — data.

instruction is a binary sequence which is designed inside the processor to perform some task.

binary sequence — bind with — operation

Data is a binary sequence which is associated with a value based on positional weights of number system.

binary seq — bind with — value

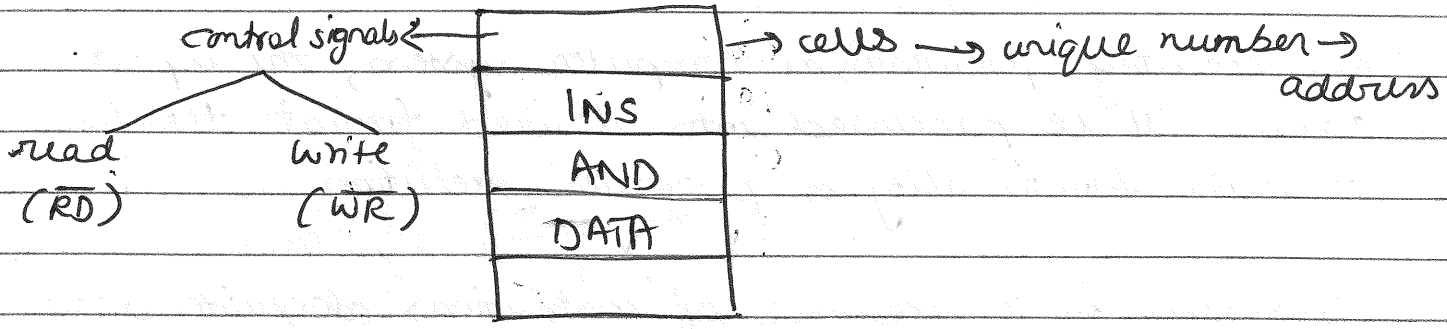
Computer system consist of fundamental components used to execute program named as CPU
- CPU (processing)
- memory (storage)
- I/O (external communication)

memory organisation

memory is storage element, so the program is always present in memory.

memory chip is divided into equal parts called as cells. Each cell is identified with a unique no called address.

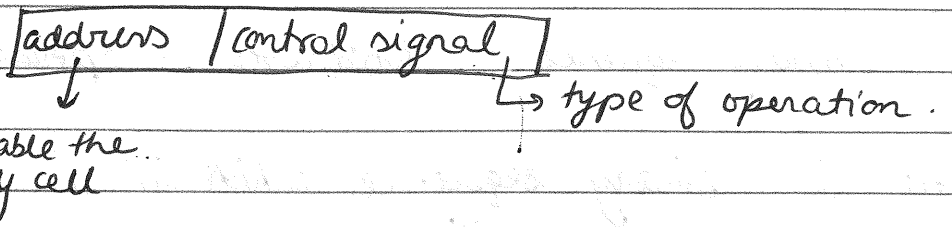
memory chip recognises the control signal to indicate type of operation.



Memory chip

CPU generates memory request to access the program

Memory request consist of two parameters



memory chip organisation

no. of address bits required to access the memory always depends on capacity of memory chip

memory chip configuration is represented as
64K X 8, 64K X 16, 64K X 32

Dt.: Pg.: Delta

eg: 64K x 8 ← bits stored in cell.

no. of cells in memory chip

64K cells → $2^6 \times 2^{10}$

$= 2^{16}$ cells.

no. of address bits required to access the memory chip

2^{16} cells ⇒ 16 bit address

address space / addr range / memory map of memory chip

$$\begin{pmatrix} 0000 & 0000 & 0000 & 0000 \\ 1111 & 1111 & 1111 & 1111 \end{pmatrix}_2$$

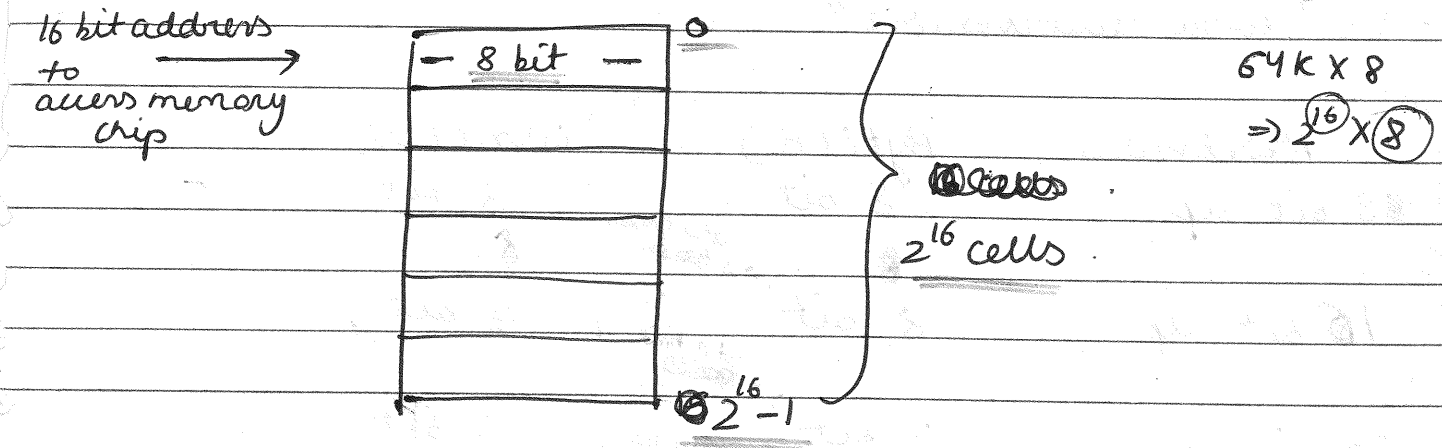
bcz of limited human recognition memory, binary address space is represented into hexadecimal addr space)

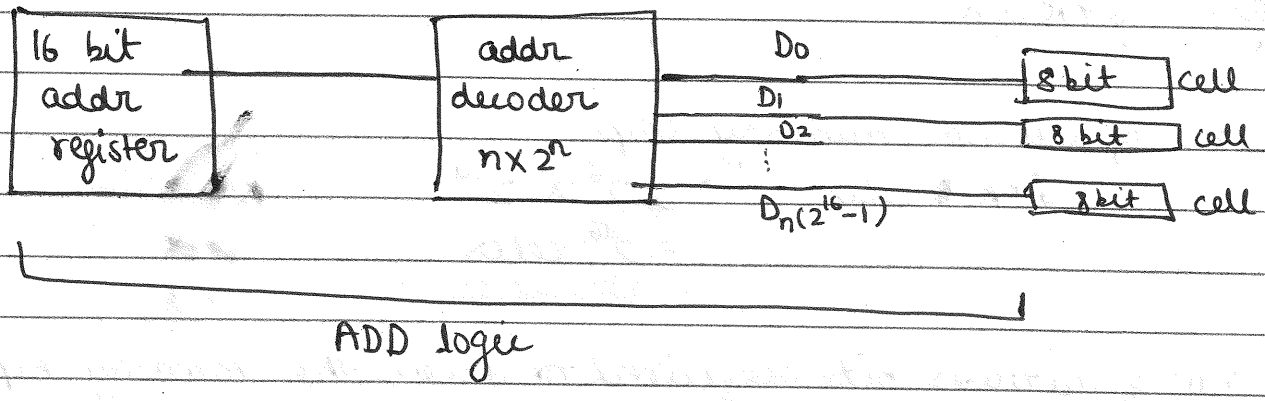
$$\begin{pmatrix} 0000 \\ FFFF \end{pmatrix}$$

16 / H / 0X

↓

(Assembler) notation.





Byte addresable VS word addresable

based on cell size of memory chip, memory configuration is divided into two types

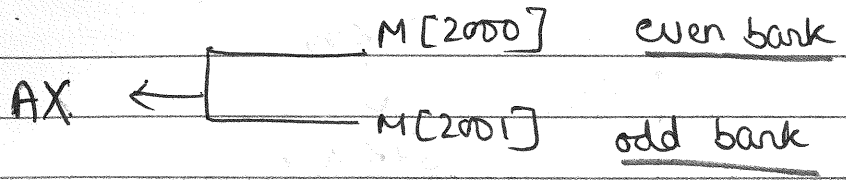
(i) **byte addresable memory** –
when memory cell size is 8 bit then corresponding address space is called "byte addresable"

(ii) **word addresable memory** –
when memory cell size depends on word length of processor then corresponding address space is called "word addresable"

Processor	Byte (B)	Word (W)
8 bit up	8 bit	8 bit
16 bit up	8 bit → 2 at a time	16 bit
32 bit up	8 bit → 4 at a time	32 bit
n bit up	8 bit → n/8 at a time	n bit

→ unique meaning
→ unambiguity
→ success

→ multiple meaning
→ ambiguity
→ unsuccessful



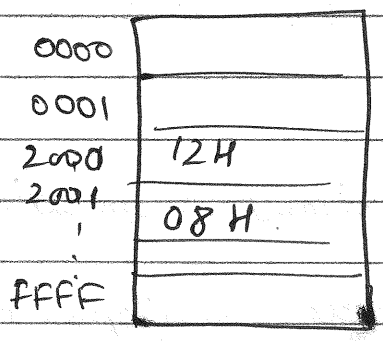
(Interface complexity is there in word length up)

AX : 15 14 . . . . 8 | 7 6 0

HB | LB

while executing the above instructions, produces the two possible outputs due to lack of order of data storage in memory that is

mem contents



AX —— M[2000]
—— M[2001]
M[2002]
M[2003]

If [2000] contains 2B then

AX = 0812H

If [2000] contains HB then

AX = 1208 H

To handle above problem, memory address interpretation mechanism (ENDIAN-NESS mechanism) are used

↓

shows order of data storage in memory

Dt.:
Pg.:

eg: 64 K·B.
↓
64K X B
↓
64K X 8

eg: 64K W
↓
64K X W
↓
64K X ?

default memory configuration in processor's design is byte addressable.

When the word length of processor is greater than 8 bit then multiple cell accessing is required to (read or write) one word data (from/to) the memory chip.

eg: 16 Bit up [8086]

MOV	AX	[2000]

MOV → mnemonic user understandable format of operation.
↓
opcode - machine understandable code of operation.
↓
type of operation (data transfer)

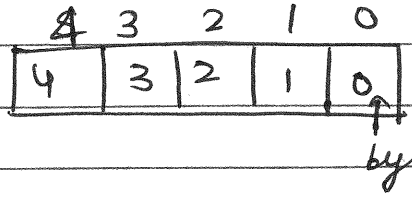
AX - destination register (register is referred with a name)
└ 16 bit

[2000] → source → memory → [memory is referred with a address] → [2000] → byte addr.

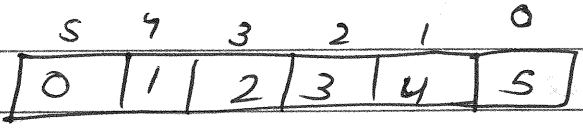
Dt.:
Pg.:

There are two kinds

(I) Little ENDIAN - Lower address contains lower byte and higher address contains higher byte.

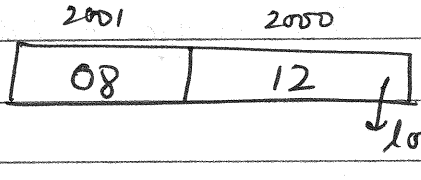


(II) BIG ENDIAN - Lower address contains higher byte and higher address contains lower byte.



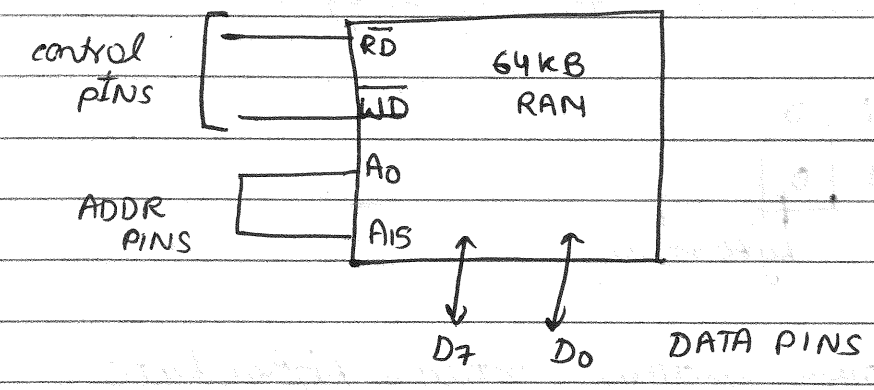
most of processor's designer uses Little ENDIAN mechanism as default address interpretation technique so above eq^n generates following output:

eg: AX:



AX: 0812H output fixed.

RAM chip design



PINS Structure.

processor consists of set of hardware pins to perform the operation on memory and I/O which are externally connected to processor.

(i) Active low pins: are enabled when clock pulse is in low state. This is denoted as $\overline{\text{Pinname}}$.

eg: $\overline{RD}$, $\overline{WD}$, $\overline{INTA}$

(ii) Active High pins: are enabled when clock pulse is in high state. It is denoted by Pinname.

eg: ALE, INTR, HOLD, HLDA

(iii) Time multiplexed pins: are used to carry the different meanings but not at same time.
On processor's design data pins are time multiplexed with address pins that means same address pins is also used to carry the data therefore advantage of time multiplexing reduces hardware pins in CPU.

addr → (2^{16}) ↓ 64KB X 8 bit ↑ ← data

eg: In 8085 up [AD0 – AD7]
addr / data

In 8086 up
↓ 1MB X 16 bit
2^{20} addr ; ↓ data

[AD0 – AD15] ↓ data

[AD0 – AD20] ↓ addr

ALE pin : is required to differentiate meaning of time multiplexed pins. that is

ALE → 0 : time multiplexed pins (carries data)

ALE → 1 : time multiplexed pins (carries address)

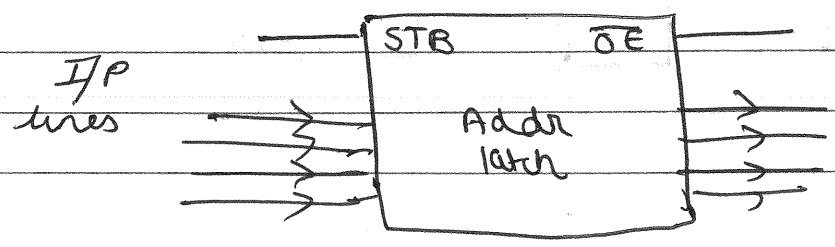
Address latch – it is temporary storage component used to hold the address. It consist of two control pins (STB and $\overline{OE}$) used to control the I/O lines of latch respectively.

STB pins → 0 : I/P lines are disabled

STB pins → 1 : I/P lines are enables

$\overline{OE}$ pins → 0 : O/P lines are enabled

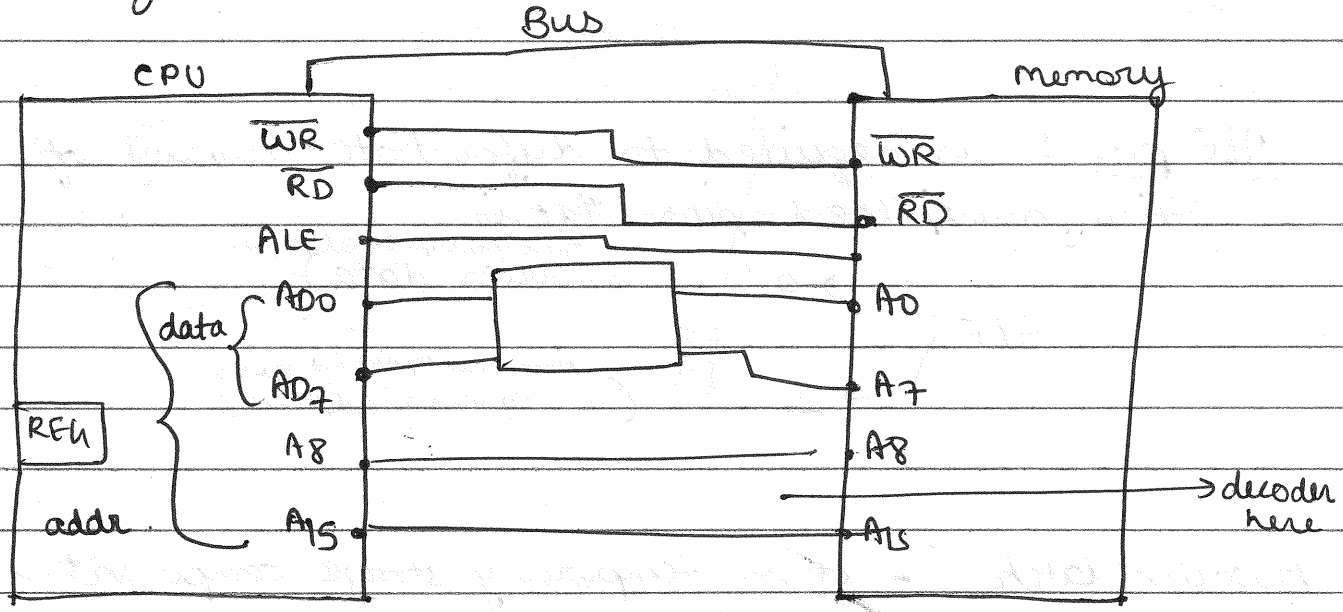
$\overline{OE}$ pins → 1 : O/P lines are disabled



Memory Interfacing

To analyse the memory management interfacing concept, consider 8085 as reference CPU.

It is a 8 bit microprocessor that supports 64 KB physical memory.



eg: MOV r0, [2000]

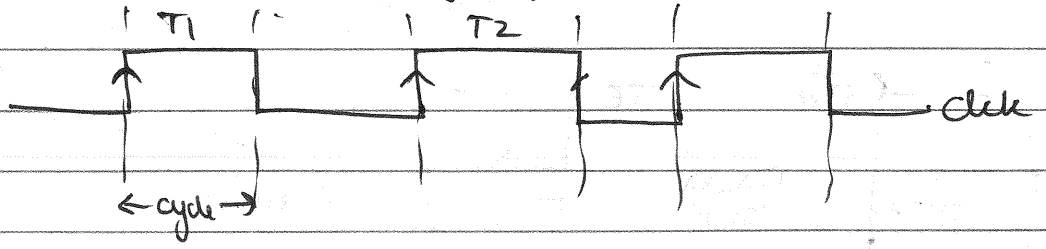
⇓

r0 ← m[2000] CPU generates memory request

[2000 & $\overline{RD}$]

machine cycle {T1, T2, T3, T4}

CPU cycle to perform the req of memory operation = memory cycle.

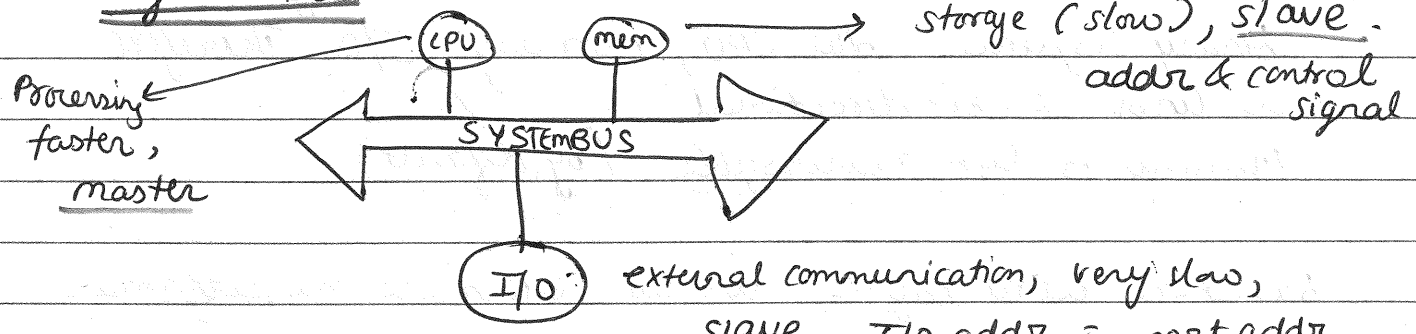


Two memory banks $D_0 - D_7$ $D_8 - D_{15}$

odd bank — even bank.

BUS

It is a communication channel. The characteristic of bus is shared transmission medium. The limitation of bus is only one transmission at a time. The bus which is used to provide communication b/w CPU, memory and I/O called the system bus.



Flip flop latency is very low.
ALU — Gate — Gate Delay is low

I/O electromagnetic components data transfer is very low.
mem → DRAM → slow

System bus : consists of three categories of line used to provide effient communication b/w CPU, memory, and I/O. These three categories are.

(1) Addr Bus — are used to carry address towards memory & I/O. They are unidirectional. Based on addresslines, we can determine the capacity of memory system.

Dt.: Pg.: Delta

eg: In 8085, 16 addr lines are present

$AD_0 - AD_7$ $AD_8 - AD_{15}$

⇓

2^{16} cells.

⇓

64 K cells.

⇓

64 KB.

(ii) Data bus — These lines are used to carry ~~dat~~ data/ binary sequence b/w CPU, memory, I/O. Therefore these lines are bi-directional

Memory is byte addressable. (by default)

based on word length we can determine the performance of processor

64 Bitup ⇒ 64 word length.

according to word length, memory interfacing depends on

eg: In 8085, 8 data pins are present AD_0 to AD_{15}

⇓

8 bit operation are performed.

(iii) Control lines/Bus — These lines are used to carry control signal and timing signal.

CPU contains clock which control the memory operation and I/O operations with CPU clock.

we cannot say that they are unidirectional/ bidirectional. They are ON/OUT ie IN unidirectional/ OUT unidirectional.

control signals are used to indicate the type of operation whereas timing signals are used to synchronize the memory and I/O operation with Processor clock.

Note: When there is common bus and common control signal, common address space used b/w memory and I/O then there is possibility of ambiguity.

Bus configuration are used in system design.

Three different configuration are used in computer design

(i) IOP [Input - Output - Processor]
common control signal, common addr space, but different buses for both memory & I/O.

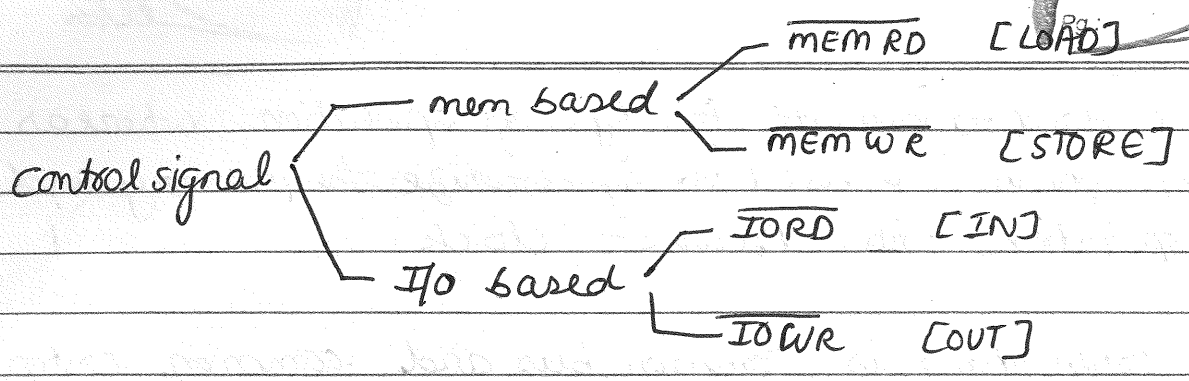
common bus	common control signal	common addr space
X	✓	✓

additional control logic is required to manage memory bus and I/O bus. Therefore it is expensive process. Therefore it is used in high performance CPU Design

(ii) Isolated I/O
same common bus, common address bus but different control signal for mem & I/O.

common bus	common control signal	common addr space
✓	X	✓

Dt.: Pg.: Delta



No ambiguity.

When processor supports IN and OUT instructions as Priviliged Instructions (MUL is not privilged instructions) then we can conclude Isolated I/O configuration is used in processor

In 8085 and 8086 both of processor uses Isolated I/O configuration bcz [IN] and [OUT] instructions are Priviliged Instructions.

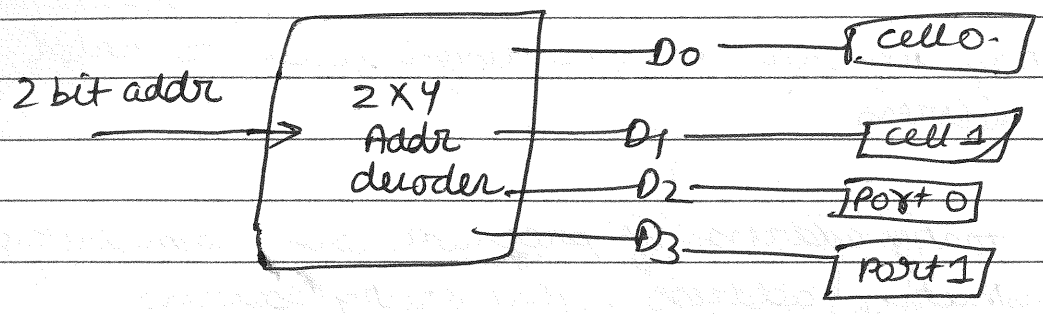
IO/M	$\overline{RD}$	$\overline{WR}$	
0	0	1	$\overline{MEMRD}$
0	1	0	$\overline{MEMWR}$
1	0	1	$\overline{IORD}$
1	1	0	$\overline{IOWR}$

(iii) Memory mapped I/O.

common bus	common control signal	common addr space.
✓	✓	✗

This configuration has common bus and common control signal but unique address space for both memory & I/O. Unique address space is maintained by taking some of the memory address from memory space and assign them to I/O ports. Disable the same address in memory space

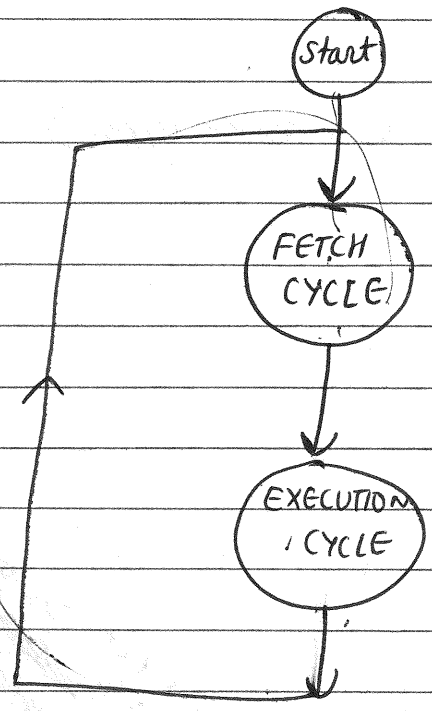
In this configuration, I/O address becomes part of memory address space, therefore limitation of configuration is complete memory space is not in use.



INSTRUCTION CYCLE

This concept describes the execution sequence of program. It consist of two subcycle.

Fetch cycle — Execution cycle



HALT : unconditional jump to some location

→ L : jmp L

Dt.:
Pg.:

→ The objective of fetch cycle is Instruction fetch. In this process CPU reads the Instruction from the memory based on PC (program counter)

→ PC is mandatory register in CPU design used to hold the Ins address.

→ PC holds the starting address of program and immediately points to next instruction address. Here starting address is supplied by the programmer and next instruction address is calculated by incrementing the PC with a step size.

→ Step size depends on the length of Instruction.

→ Finally PC holds the address of Instructions during the execution of current instructions.

INS FETCH (IF)

8085, 8086 ⟶ CISC TYPE (variable type)

(I) 1 byte instruction.

SIM
RIM

(II) 2 byte instruction.

MVI 23H

(III) 3 byte instruction

LDA 2080H.

↓

LDA 0010 0000 1000 0000

↑ 3rd memory mai store hoga.

↑ 2nd memory mai store hoga

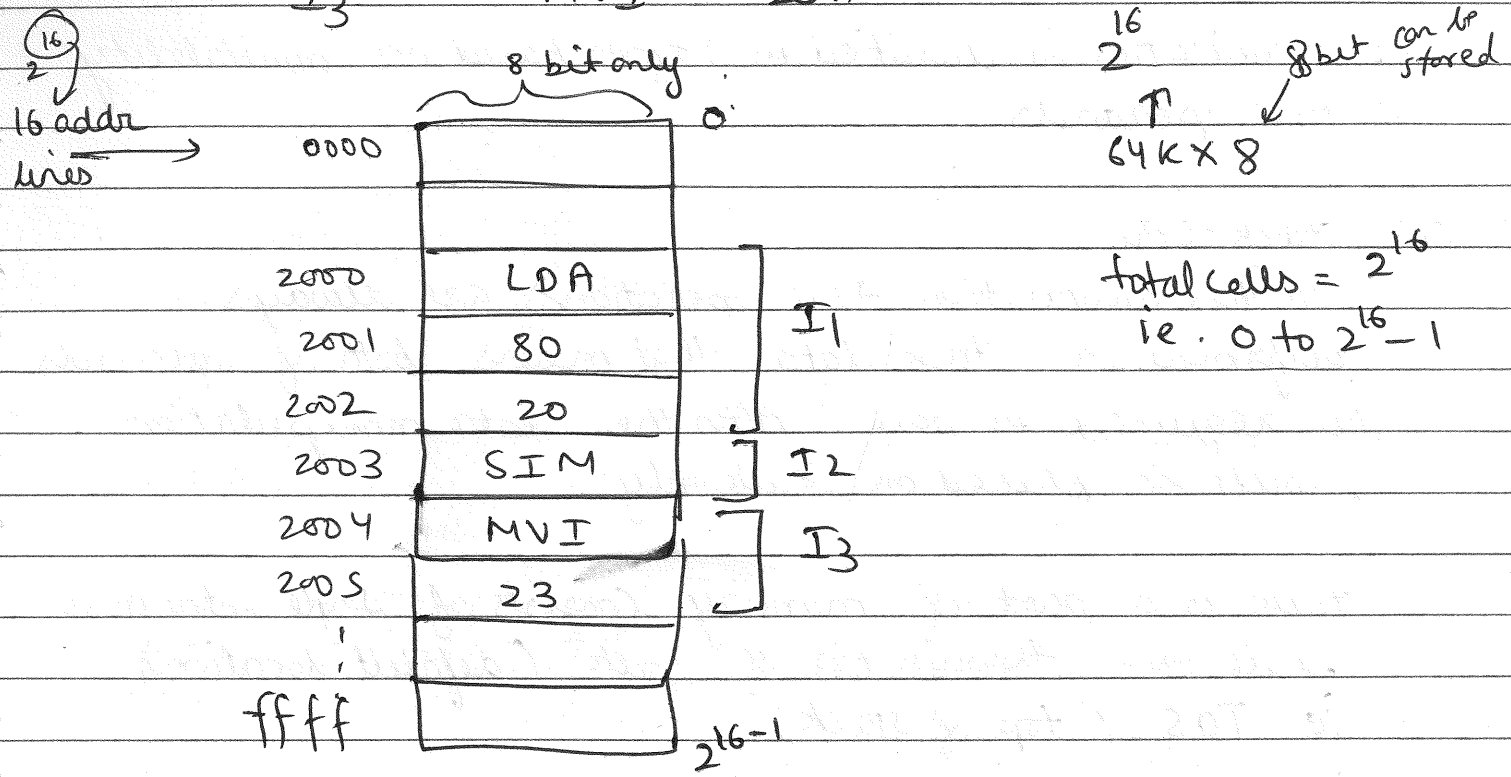
↑ 1st memory mai store hoga

8 bit can be stored only to one memory space.

I_1 LDA 2080H

I_2 SIM

I_3 MVI 23H



PC is a referential element of CPU during instruction fetch cycle.

Execution cycle.

The objective of execution cycle is processing of currently fetched Instruction. OPcode is required to execute the instruction. OPcode is part of Instruction, specified by Instruction format.

INS FORMAT

Instruction format describes the Internal structure of Instructions (layout design) of Instruction.